

RetailMind Analytics — Business Report

RetailMind Analytics — Automated Business Report

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Executive Summary

- Total revenue: £7,405,544.45
- Total orders: 18,386
- Total customers: 4,333
- Average order value: £402.78
- Highest revenue country: United Kingdom
- Top-selling product by quantity: MEDIUM CERAMIC TOP STORAGE JAR

Business Insights

- Customer segmentation helps identify high-value, loyal, and at-risk customers.
- Market basket rules reveal products that can be bundled or cross-sold together.
- Anomaly detection highlights unusual invoices that may need manual business review.
- Time-series analysis supports planning around sales seasonality and demand peaks.

Limitations

- Results are based on historical transactions and should support, not replace, human business judgment.
- Anomaly detection is unsupervised and does not prove fraud; it flags unusual patterns for review.
- Recommendations work better for customers with sufficient purchase history.

Preprocessing Summary

Original rows: 541,909

Final rows: 389,921

Total dropped: 151,988

Retention rate: 72.0%

Dataset Mode

Customer-summary mode: False

Column Mapping

- No column mapping was needed or detected.

Cleaning Actions

- Dropped duplicates: 5,268
- Dropped missing descriptions: 1,454
- Dropped missing CustomerID: 133,583
- Dropped cancelled/invalid invoices: 8,872
- Dropped non-product stockcodes: 1,549
- Dropped invalid quantity/price: 33
- Dropped invalid dates: 0
- Dropped rare products: 1,229

Feature Engineering

- TotalPrice = Quantity × UnitPrice or loaded directly from Sales/Amount column
- Month, MonthName, DayOfWeek, DayOfWeekNum, Hour, IsWeekend, YearMonth
- TotalPrice was derived from Quantity × UnitPrice.

Outlier Handling

- TotalPrice was clipped using 1% and 99% quantiles.
- Lower cap: 0.55
- Upper cap: 201.6